# Vakint reduction benchmark

Cached RustRed versus the FORM / FMFT reduction route

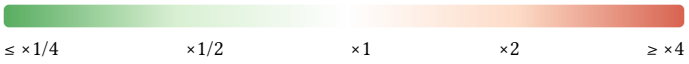
2026-10-20 · AMD EPYC 9754 128-Core Processor · CPU affinity 190

**Reading guide.** Bold values are RustRed times; FMFT times appear below. Multipliers are *RustRed ÷ FMFT*: green means faster, white parity, red slower. Both receive identical four-loop scalar inputs. The first physical denominator has power 2 (base) or 4 (held out); all others have power 1, auxiliaries 0.

## Reduction time

Median seconds over 3 fresh processes per backend and case.

| Topology | Base · power 2 | Held out · power 4 |
|---|---|---|
| H | **0.0474 s**<br>FMFT 0.639 s ×0.074 | **3.21 s**<br>FMFT 7.22 s ×0.45 |
| FG | **0.0294 s**<br>FMFT 0.166 s ×0.18 | **0.0821 s**<br>FMFT 0.347 s ×0.24 |
| BMW | **0.0126 s**<br>FMFT 0.123 s ×0.10 | **0.721 s**<br>FMFT 3.11 s ×0.23 |
| X | **0.0888 s**<br>FMFT 1.55 s ×0.057 | **13.0 s**<br>FMFT 21.3 s ×0.61 |




**Method and correctness.** Release build: optimization level 3, fat LTO; rustc 1.98.1; FORM 4.3. Both retain masters in FMFT/MATAD normalization, with identical symbolic mass. Exact rational parity passed all eight cases at unit mass and renormalization scale, without truncating epsilon or using numerical tolerances.

Base cases hit prepared roots without applying rules. Held-out cases reuse cached descendants after root misses. Timing covers first public reduction calls, including FORM version checks, launch, I/O and parsing. Data loading, cache attachment, master evaluation and validation are excluded. Fresh processes run sequentially, alternating backends, with the recorded CPU affinity on a shared host. Filesystem pages are warm; this does not measure cold startup.

Measured source: `624bfd919313`. Modified source tree. Binary: `f6cb28c1c51a`. Data: `v0.1.0`. Samples and provenance: `data/results.json`. Refresh instructions: `../../AGENTS.md`.